VoiceRun: Voice-Powered Running App

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Abstract

The problem

- While running, it’s common to keep track of the **progress** using an application

- Many runners have found the interaction with such apps inconvenient, often requiring you to **physically touch** the screen

- To properly **check** the current information or to end the current session, **stopping** becomes almost mandatory

The solution

- Using a **voice user interface** we can avoid the pitfalls of touch-based interface

- By using voice commands, users can keep their full attention on the activity **without interruptions**

Abstract

- About the project
- Tasks & Storyboard
- Prototype
- Application
- Conclusion

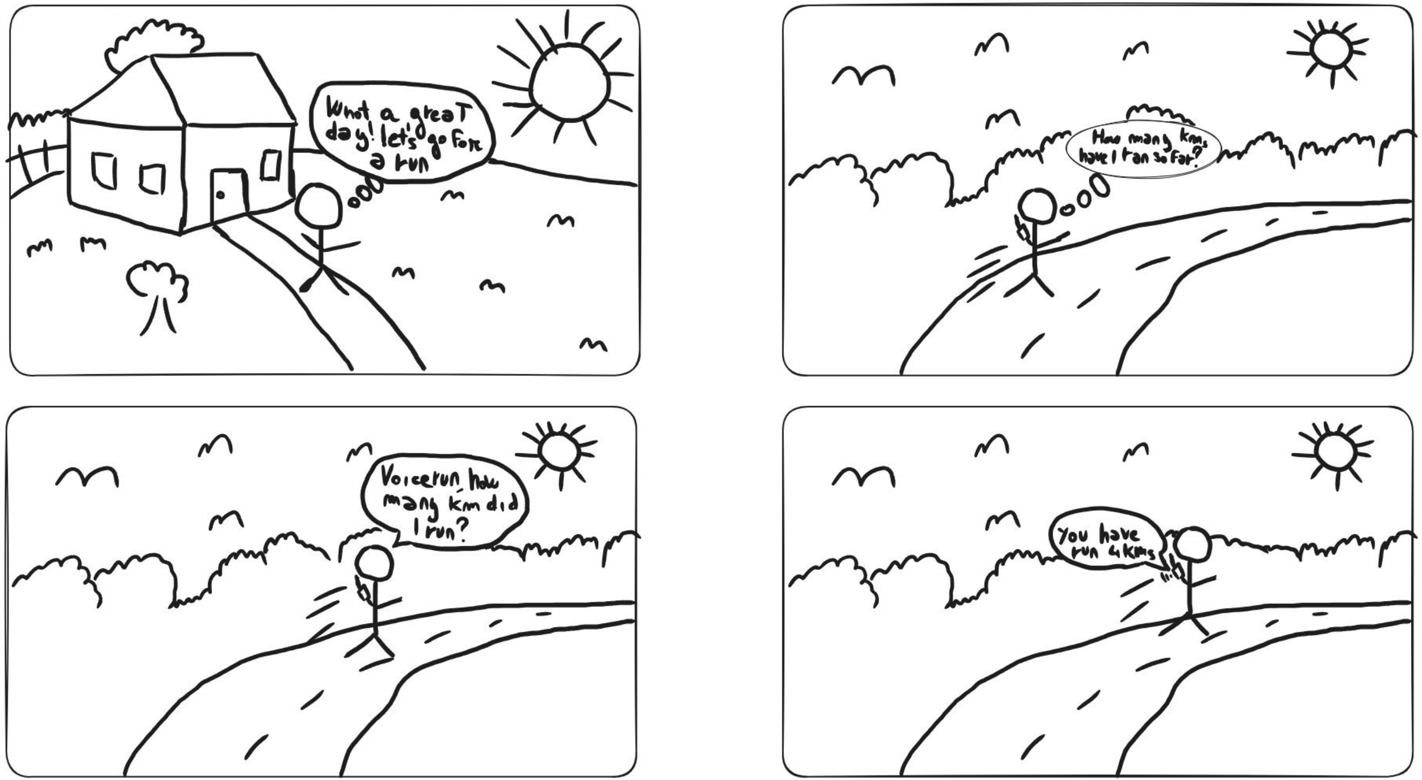
Tasks

- The user wants to know:
  - their current **pace** mid-run
  - the current **distance** mid-run
  - the **time elapsed** since the beginning of the run
  - the **calories** burned during the run

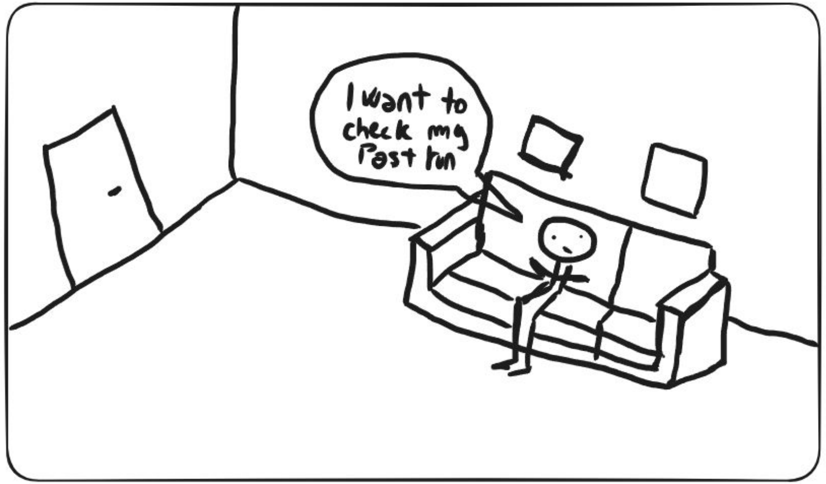
- The user wants to view the **history** of their past runs

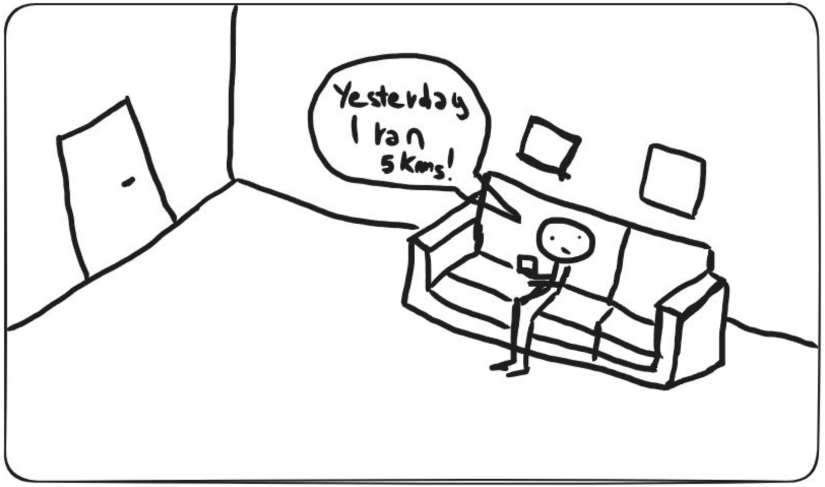
- The user wants to control the running session by **pausing**, **resuming**, or **stopping it**

Storyboard - Vocal Interaction



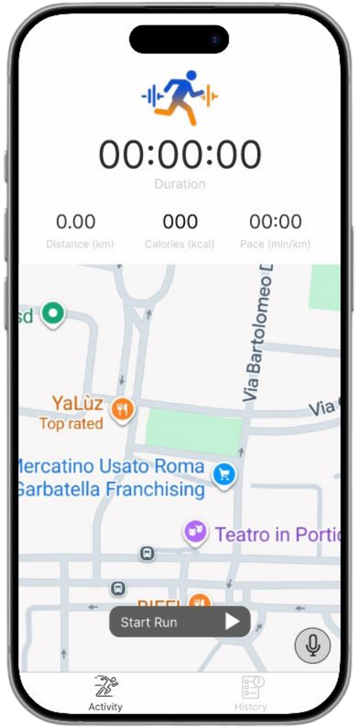
Storyboard - Run History

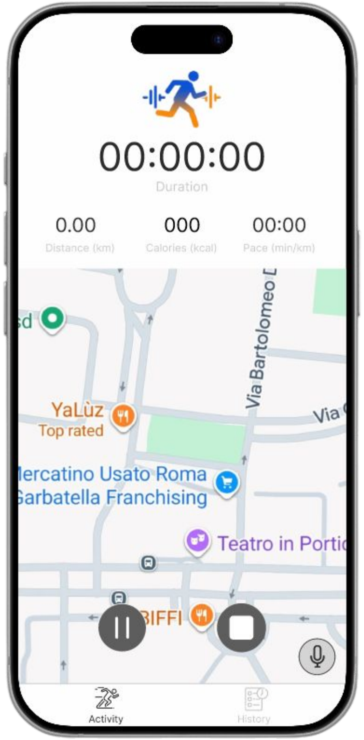




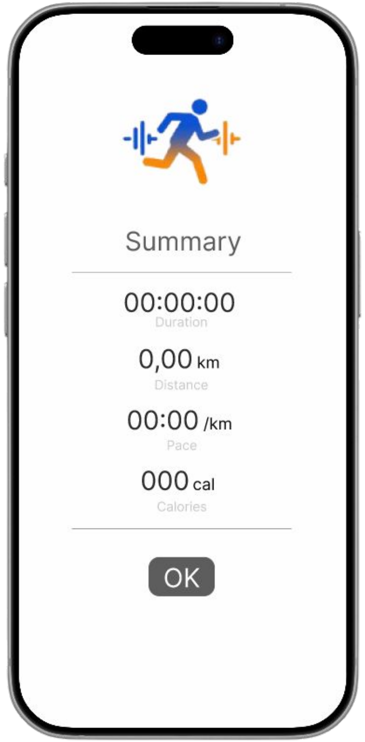
Abstract

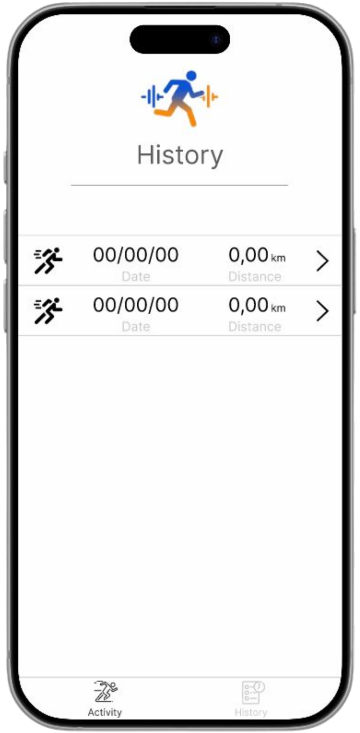
Prototype - Figma





Prototype - Figma





Abstract

React Native & Expo

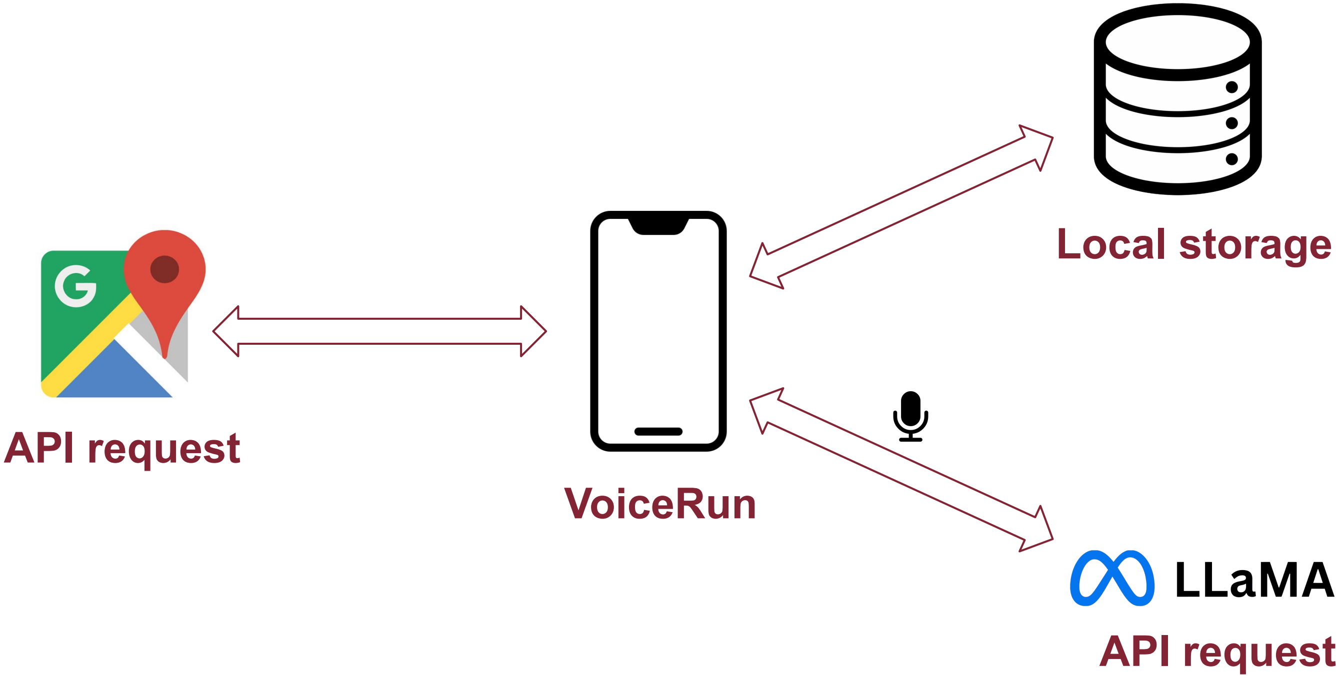
- The framework used to build our application is **React Native**
- We also used **Expo**, a **full-stack** framework to smoothen the development process of the app.
- Thanks to Expo it was possible to build the app for **both** Android and iOS.

Llama 3.1

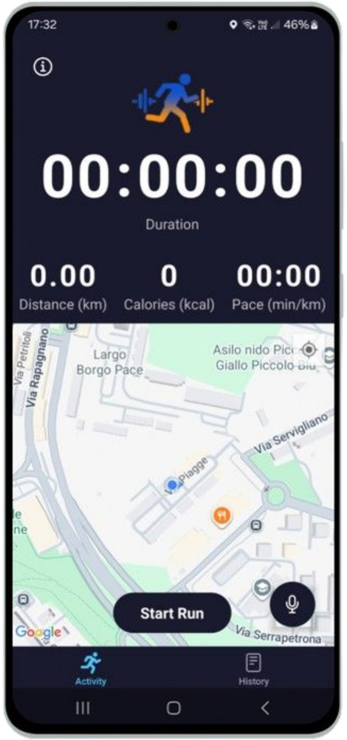
- We used **Groq** for its fast inference time, while the actual language processing was handled by **Llama 3.1**

MODEL	REQUESTS PER MINUTE	REQUESTS PER DAY	TOKENS PER MINUTE	TOKENS PER DAY	ACTIONS
allam-2-7b	30	7K	6K	500K	
groq/compound	30	250	70K	No limit	
groq/compound-mini	30	250	70K	No limit	
llama-3.1-8b-instant	30	14.4K	6K	500K	
llama-3.3-70b-versatile	30	1K	12K	100K	
meta-llama/llama-4-scout-17b-16e-instruct	30	1K	30K	500K	
meta-llama/llama-prompt-guard-2-22m	30	14.4K	15K	500K	
meta-llama/llama-prompt-guard-2-86m	30	14.4K	15K	500K	
openai/gpt-oss-120b	30	1K	8K	200K	

App Structure



Activity Tab

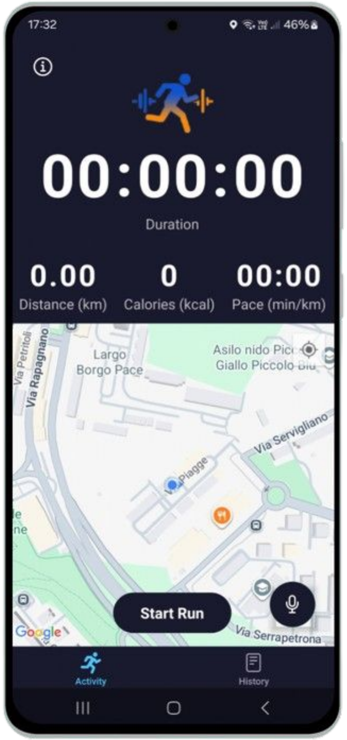


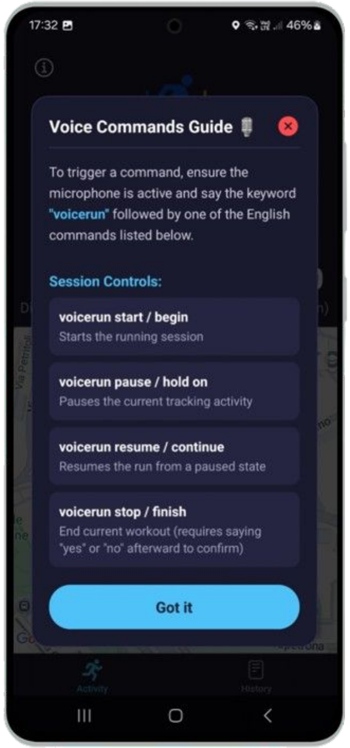
Start an activity, then **stop/resume** or **end** it

Ending an activity shows its **summary**

You can use vocal commands to make the app talk about requested information (**pace,km,time**,ecc…)

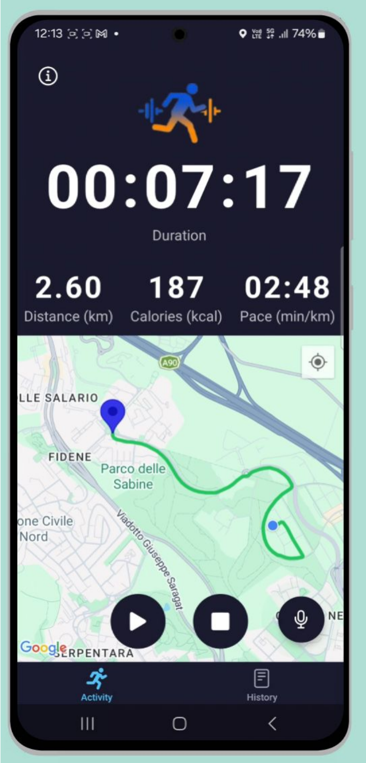
Activity Tab - Commands Guide

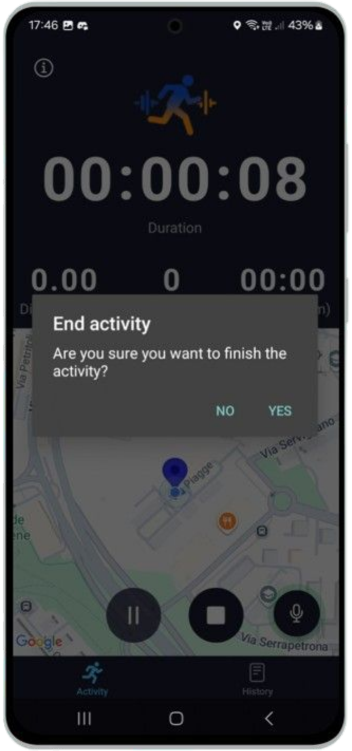


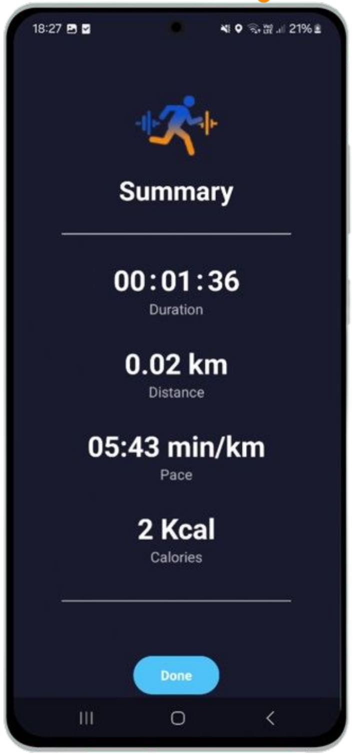


Application ○○○○○○●○○○○○○

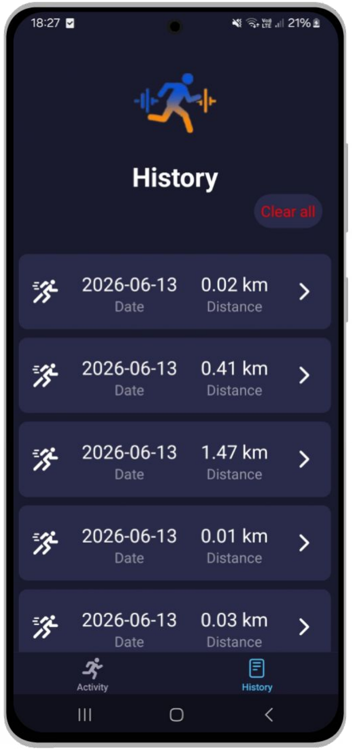
Activity Tab - While Running







History Tab

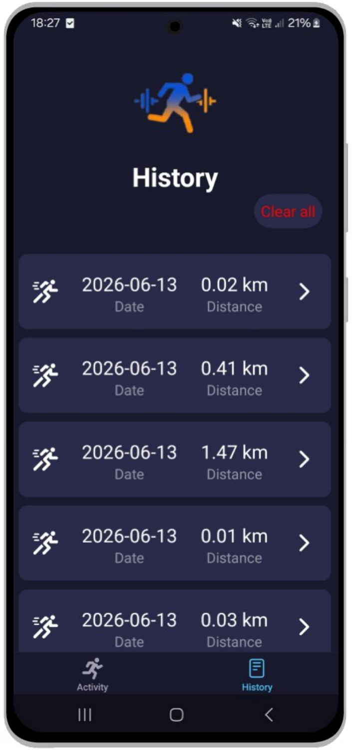


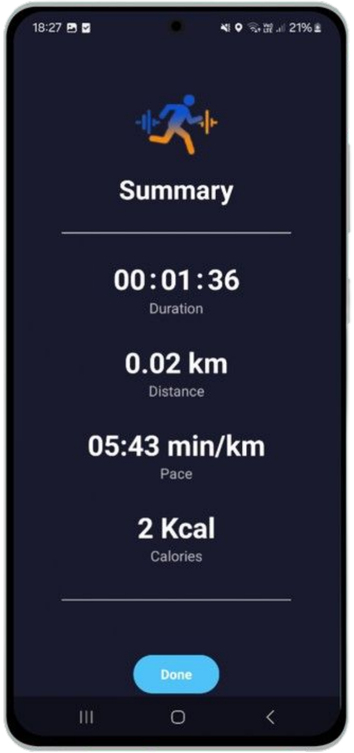
Check past runs

Click on a run to access its **summary**

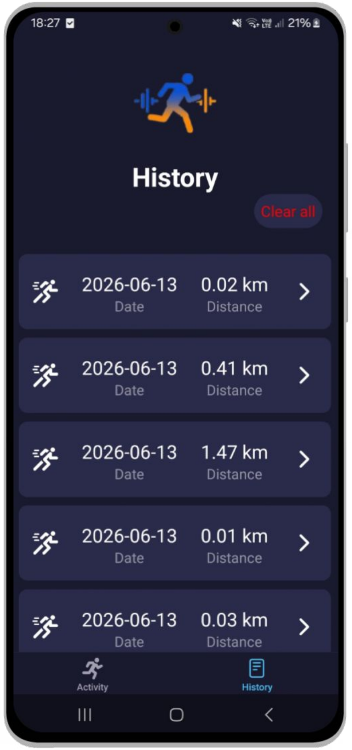
Delete all runs

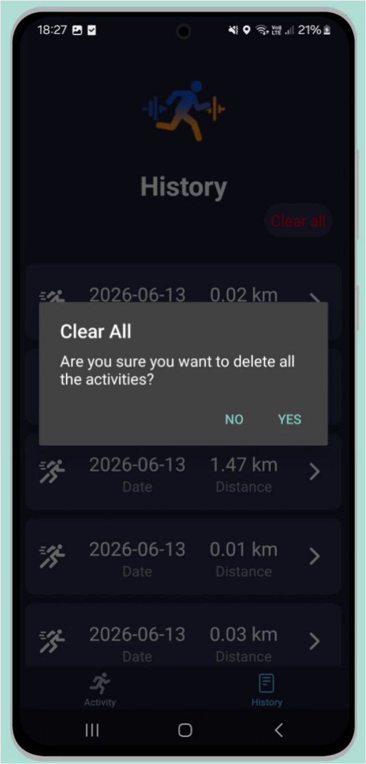
History Tab - Summary

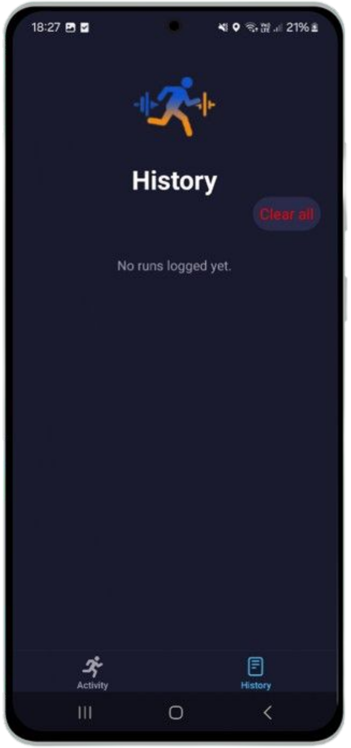




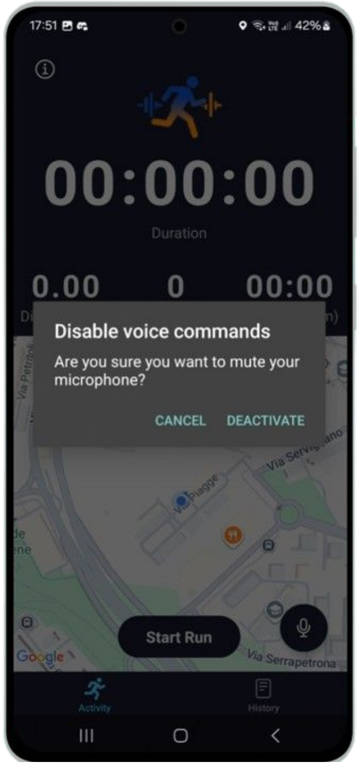
History Tab - Clear All

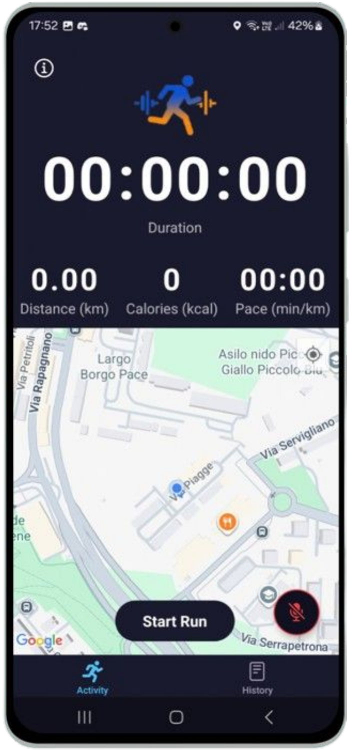




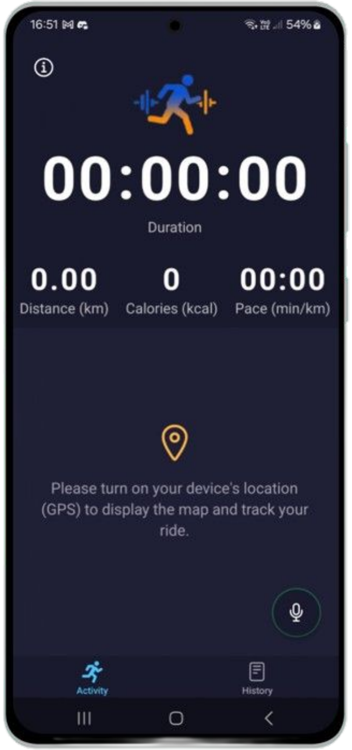


Edge Cases - Disable Microphone

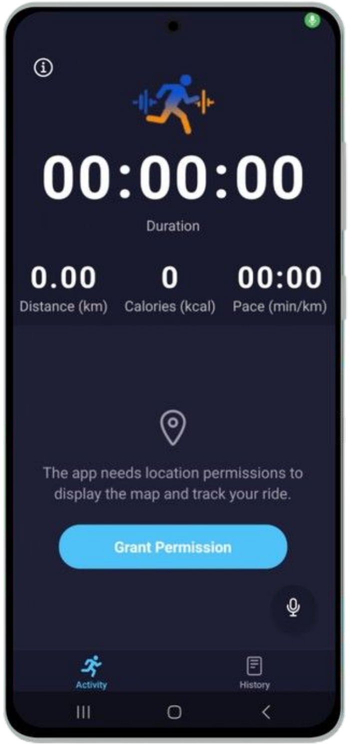




Edge Cases - Turn Off Location



Edge Cases - Location Permissions



Abstract

Thoughts and future developments

- VoiceRun performs its tasks **well**, allowing users to use their voice for interactions

- For possible future developments, **increasing the receptivity** and understanding of the vocal commands could lead to a **smoother user experience**

- Beyond that, using other paid models could allow the app to actually be distributed without having to worry about **tokens limit**

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Thank you for the attention!